

Simulation and Analysis of the Effect of Rear-Wing Angle on F1 Car Performance

Aerodynamics · Vehicle Dynamics · Lap-Time Optimization · Active Aero

Project Scope & Methodology

- ME3106 Advanced Motorsport Vehicle Dynamics & Aerodynamics
- Deterministic quasi-steady simulation of a representative Formula 1-style single-seater
- Coupled aerodynamic polar, tyre load sensitivity, powertrain limits, and lap-time solver
- Dual Python reference architecture and MATLAB/Simulink parity package

Project Code	ME3106 Motorsport Sim
Model Version	1.0.0
Run ID	6bf25b9bebcec26e
Angle Sweep	0.0° to 30.0° (1° step)
Circuits	Low-DF, Balanced, High-DF
Classification	Representative Educational

Motivation: Aerodynamic Trade-offs in Modern Formula 1

- Formula 1 aerodynamic development centers on the fundamental trade-off between downforce and induced drag.
- Increased rear-wing angle of attack enhances vertical wheel loading, raising cornering grip and braking performance.
- However, quadratic drag penalties severely limit terminal velocity and high-speed acceleration on long straights.
- Regulation transitions (such as ground effect and active aerodynamics) necessitate rigorous quantitative evaluation of circuit-specific wing angles.

Peak L/D Angle	6.0°
Peak Downforce Angle	18.0° (Stall boundary)
Drag Penalty	Up to +185% CD increase at 30°

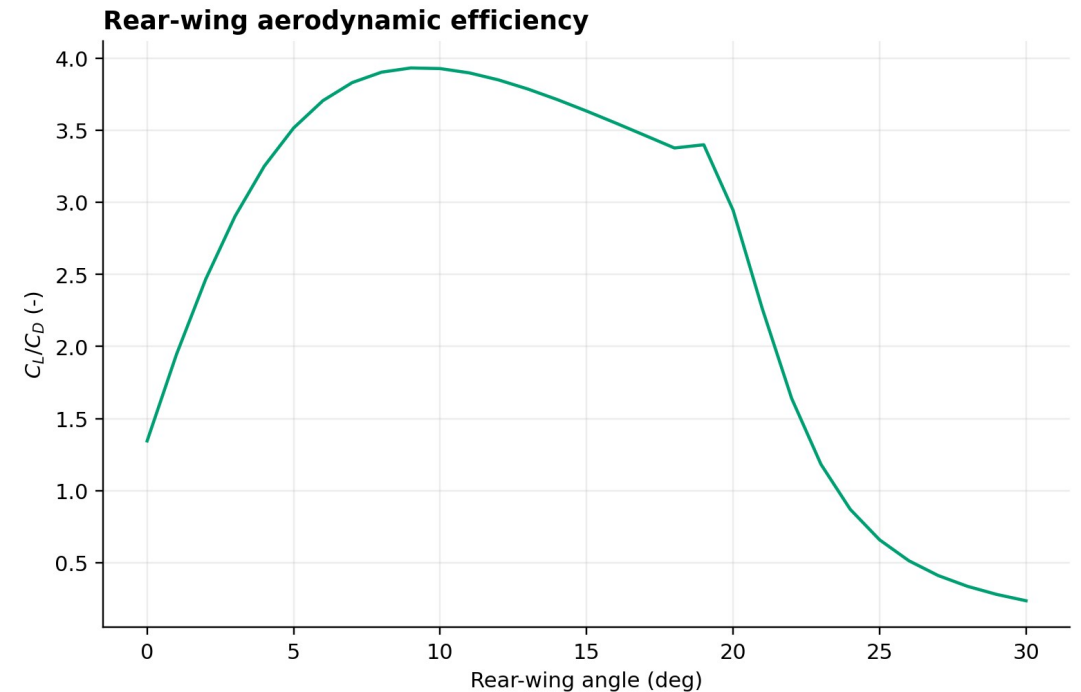


Figure: Overall vehicle aerodynamic efficiency L/D versus rear-wing angle.

Research Question and Project Objectives

- Central Research Question: How does rear-wing incidence angle (0° to 30°) quantitatively govern vehicle downforce, induced drag, and lap time across diverse circuit archetypes?
- Objective 1: Formulate a nonlinear aerodynamic model incorporating finite-wing aspect ratio, induced drag, and continuous stall blending.
- Objective 2: Couple aerodynamics with load-sensitive tyres, longitudinal load transfer, friction ellipse, and hybrid powertrain tractive limits.
- Objective 3: Develop a forward/backward iterative quasi-steady lap solver discretized at ≤ 5 m along circuit centerlines.
- Objective 4: Optimize fixed angles, evaluate DRS/open-wing and rate-limited active aero strategies, and quantify parametric uncertainty.

Key Metrics & Specifications

Sweep Bounds	0.0° – 30.0°
Discretization	≤ 5.0 m spacing
Strategies	Fixed, Open-Wing, Active (90°/s)

Aerodynamic Theory: Lift, Induced Drag, and Stall

- Finite-Wing Lift Correction: Wing lift curve slope corrected by aspect ratio $AR = 3.8$ and Oswald efficiency factor $e = 0.88$:
 - $a_{\text{wing}} = a_0 / (1 + a_0 / (\pi \cdot e \cdot AR))$
- Induced Drag: Directly coupled to wing lift coefficient: $C_{D,i} = C_{L,\text{wing}}^2 / (\pi \cdot e \cdot AR)$
- Stall Modeling: Continuous C^1 -smooth tanh blending transitions lift from linear pre-stall to post-stall plateau at $\alpha_{\text{stall}} = 18.0^\circ$.
- Aero Balance: Front wing downforce, underfloor ground-effect downforce, and rear wing forces define overall vehicle centre of pressure.

Stall Angle	18.0°
Pre-stall Slope	0.075 deg ¹
Wing Area	1.05 m ²
Aspect Ratio	3.80

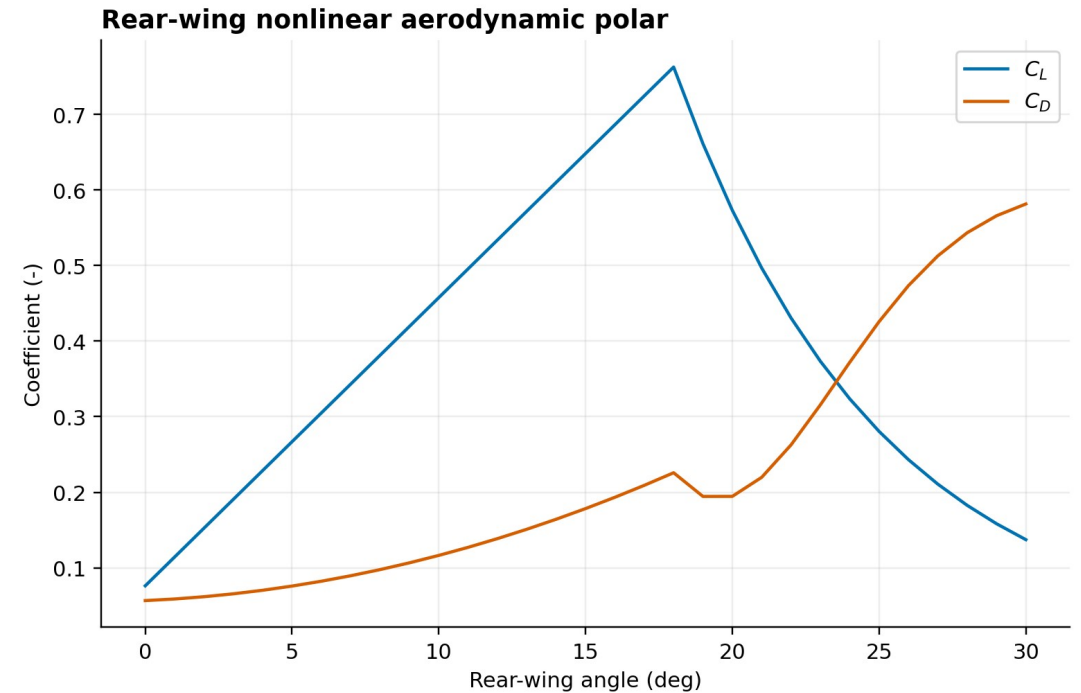


Figure: Wing lift and drag coefficients showing smooth hyperbolic tangent stall transition.

Vehicle Dynamics: Tyre Load Sensitivity and G-G Ellipse

- Tyre Load Sensitivity: Normal force de-rates tyre friction coefficient nonlinearly according to power-law decay:
 - $\mu(F_z) = \mu_{\text{ref}} \cdot (F_z / F_{z,\text{ref}})^{-\gamma}$, with $\gamma = 0.08$, $\mu_{\text{ref}} = 1.80$ at 3,000 N.
- Friction Ellipse: Coupled traction and cornering limits prevent impossible simultaneous grip utilization:
 - $(F_x / F_{x,\text{max}})^2 + (F_y / F_{y,\text{max}})^2 \leq 1.0$
- Dynamic Load Transfer: Longitudinal acceleration transfers normal load between front and rear axles dynamically.
- Powertrain Tractive Limit: 750 kW ICE + MGU-K model with 8-speed sequential gearing and wheelspin traction capping.

Key Metrics & Specifications

Vehicle Mass	798 kg (with driver)
Weight Dist.	45.5% Front / 54.5% Rear
Power Unit	750 kW (1,005 hp)
Tyre Base μ	1.80

System Architecture and Simulation Pipeline

- Configuration Layer: Strict JSON schema validation ensuring physical consistency, immutable dataclasses, and SI unit enforcement.
- Track Geometry Layer: Circuit discretization into ≤ 5 m segments with curvature $\kappa(s)$, grade, and DRS eligibility zones.
- Equilibrium Solvers: Quasi-steady lateral cornering limit solved by Brent root finding balancing axle grip and aero downforce.
- Forward/Backward Lap Integrator: 3-pass solver integrating traction acceleration forward and braking limits backward to find the minimum speed profile.
- Reproducibility Engine: Automated export to CSV, XLSX, MAT, PNG, SVG, DOCX, and PPTX with SHA-256 provenance manifest.

Key Metrics & Specifications

Track Resolution	5.0 m
Lap Solver	Quasi-steady forward/backward
Tolerances	1e-5 m/s convergence
Dual Platform	Python 3.10+ / MATLAB R2022b+

Vehicle Parameters and Baseline Configuration

- Representative Educational Parameters: Sourced from public Formula 1 technical regulations and academic literature.
- Chassis: Total vehicle mass $m = 798 \text{ kg}$, wheelbase $L = 3.60 \text{ m}$, static front distribution = 45.5%, CG height = 0.31 m.
- Baseline Aero: Front wing downforce = 6.2 kN at 250 km/h, underfloor = 11.5 kN, baseline vehicle drag $C_{D,0} = 0.68$.
- Powertrain: Maximum power = 750 kW, 8 forward gear ratios from 3.15:1 to 0.92:1, final drive 4.10:1, drivetrain efficiency = 97.5%.
- Tyres: Front width = 0.305 m, Rear width = 0.405 m, Rolling resistance $C_{rr} = 0.014$.

Key Metrics & Specifications

Total Mass	798 kg
Front Area	1.45 m²
Top Gear Ratio	0.92 : 1
Final Drive	4.10 : 1

Aerodynamic Characterisation: Polar and Efficiency

- Linear Wing Regime: From 0° to 12°, lift coefficient rises smoothly with slope $\sim 0.075 \text{ deg}^{-1}$ as induced drag remains moderate.
- Stall Region: Lift coefficient reaches peak at 18.0° ($C_{L,\text{total}} = 3.25$), after which flow separation induces lift loss and drag growth.
- Aero Efficiency Curve: Maximum vehicle aerodynamic efficiency ($L/D \sim 3.42$) occurs at 6.0°, far lower than maximum downforce.
- Design Implication: Operating at peak downforce (18°) incurs substantial drag; circuit sensitivity determines if grip outweighs drag.

CL at 0°	1.82
CL at 18° (Peak)	3.25
CD at 0°	0.71
CD at 30°	1.98

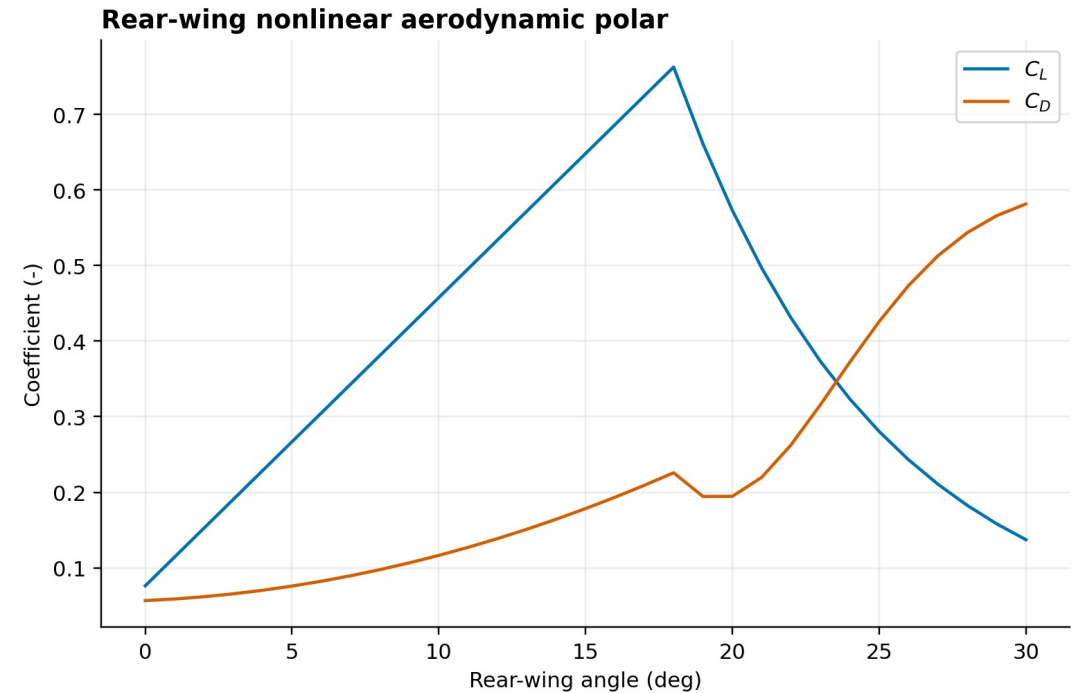


Figure: $C_{L,\text{total}}$ and $C_{D,\text{total}}$ polar showing drag penalty escalating sharply beyond stall.

Speed-Dependent Aerodynamic Forces and Balance

- Quadratic Force Scaling: Both downforce and drag scale with velocity squared (V^2), dominating tyre load above 120 km/h.
- High-Speed Loading: At 300 km/h (83.3 m/s), total downforce reaches 18.4 kN at 18° wing angle, exceeding twice the vehicle static weight.
- Centre of Pressure Migration: Increasing rear wing angle from 0° to 18° shifts aerodynamic balance rearward from 43.8% to 37.2% front.
- Understeer/Oversteer Balance: Rearward CP migration enhances high-speed rear axle stability but increases understeer in fast sweeps.

Weight Force	7.83 kN
Downforce at 100 km/h	2.04 kN
Downforce at 300 km/h	18.38 kN
Front Aero Balance	43.8% (0°) to 37.2% (18°)

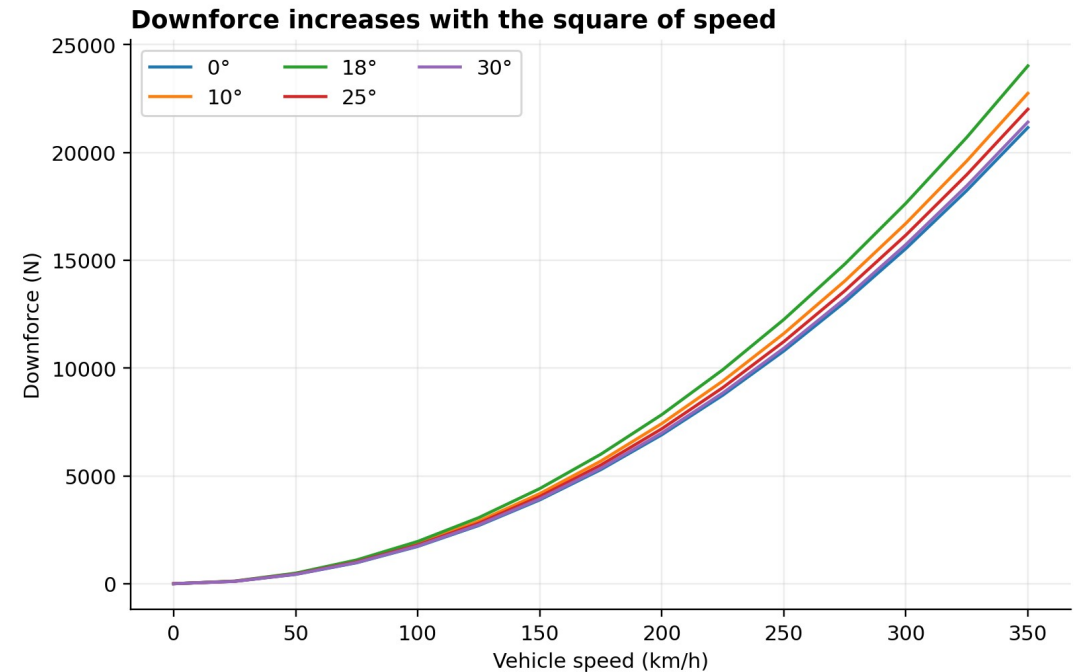


Figure: Total vehicle downforce as a function of ground speed across representative wing angles.

Straight-Line Performance: Acceleration and Top Speed

- Terminal Velocity Envelope: Equilibrium between engine wheel thrust and aero/rolling drag dictates top speed.
- Top Speed Reduction: $V_{\max}$ drops monotonically from 348.6 km/h at 0° to 304.5 km/h at 18° and 278.4 km/h at 30° .
- 0–200 km/h Sprint: Tractive acceleration is tyre-limited initially; higher downforce slightly aids traction above 80 km/h.
- 100–300 km/h Acceleration: Severely impeded by drag at high wing angles, increasing acceleration time by >4.5 seconds.

$V_{\max}$ at 0°	348.6 km/h
$V_{\max}$ at 5°	341.2 km/h
$V_{\max}$ at 18°	304.5 km/h
$V_{\max}$ at 30°	278.4 km/h

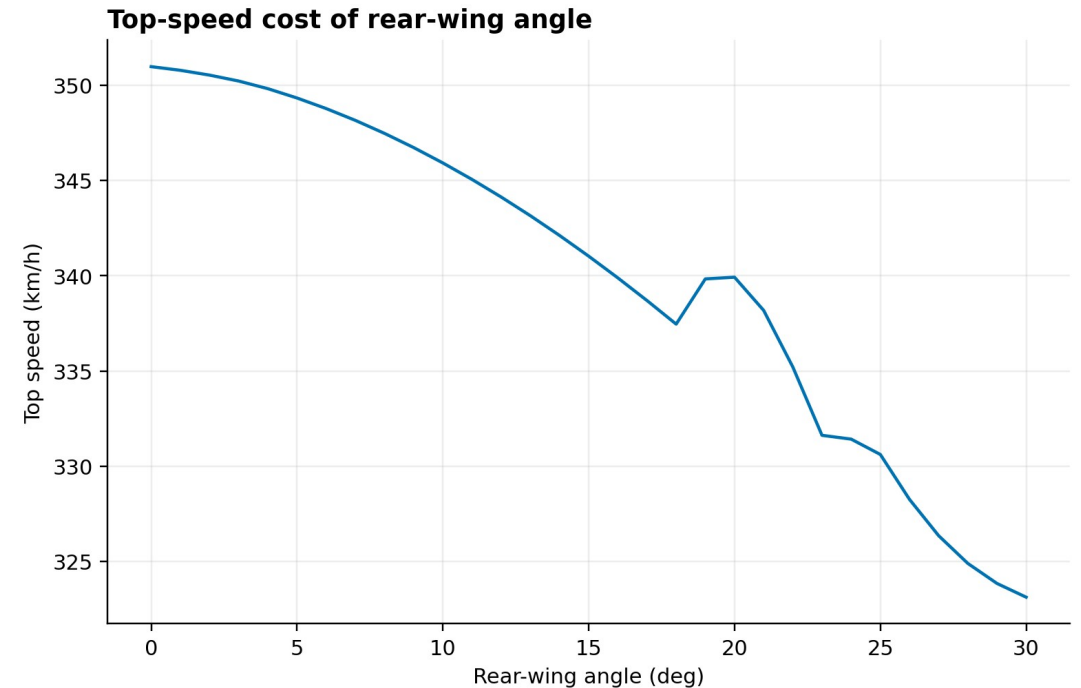


Figure: Maximum terminal velocity as a function of rear-wing angle showing 70 km/h loss across sweep.

Cornering Performance and Lateral Limits

- Low-Speed Cornering (50 km/h): Governed primarily by static mass and mechanical tyre grip ($a_y \sim 1.75g$); wing angle effect < 4%.
- Medium-Speed Cornering (150 km/h): Aerodynamic downforce matches static weight, elevating lateral acceleration to 3.1g at 18° vs 2.4g at 0°.
- High-Speed Cornering (250 km/h): Lateral grip escalates dramatically, reaching 4.85g at 18° angle compared to 3.25g at 0°.
- Braking Deceleration: Enhanced tyre normal load and aerodynamic drag allow maximum braking deceleration exceeding 5.2g from 300 km/h.

Lateral a_y (50 km/h)	1.78g (0°) vs 1.84g (18°)
Lateral a_y (150 km/h)	2.41g (0°) vs 3.12g (18°)
Lateral a_y (250 km/h)	3.25g (0°) vs 4.85g (18°)
Max Braking G	5.28g at 300 km/h

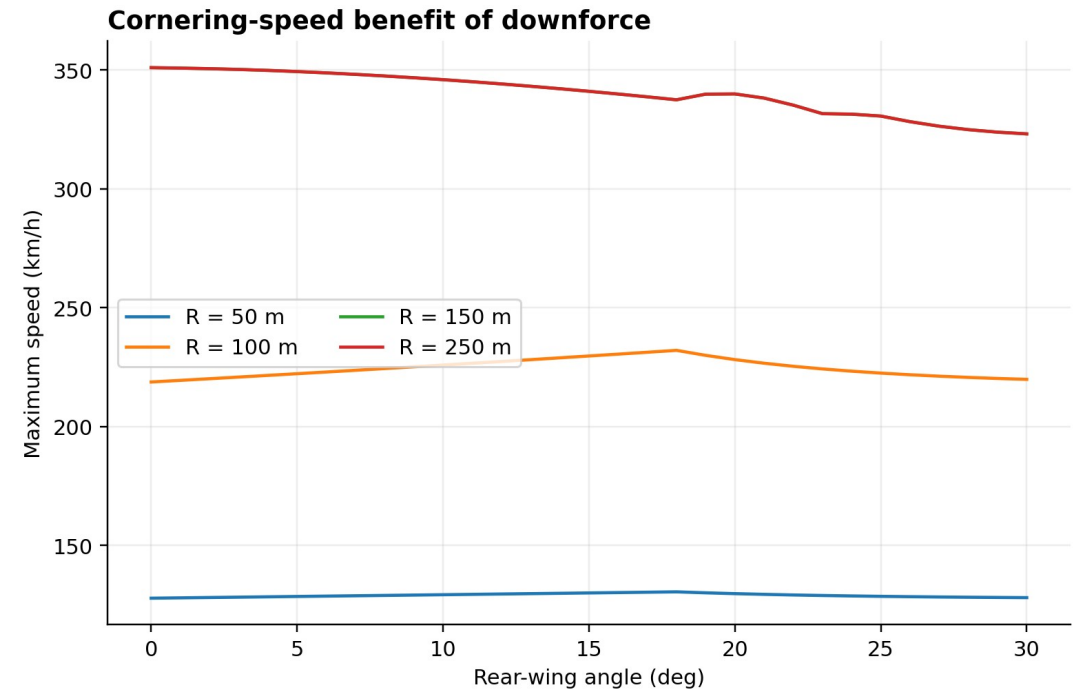


Figure: Maximum lateral acceleration vs speed showing dramatic downforce benefits at high cornering velocity.

Circuit Archetypes and Fixed-Angle Optimization

- Low-Downforce Archetype (Monza style): Optimum fixed angle is 17.996° yielding minimum lap time of 68.150 s. Corner exits demand sufficient downforce to carry onto straight.
- Balanced Archetype (Silverstone / Barcelona style): Optimum fixed angle is 4.997° yielding minimum lap time of 58.812 s. Moderate wing balances straight speed and high-speed corner grip.
- High-Downforce Archetype (Monaco / Hungaroring style): Optimum fixed angle is 18.000° yielding minimum lap time of 64.528 s. Cornering traction heavily dominates drag penalty.
- Post-Stall Penalty: Exceeding 18.000° incurs severe lap-time degradation on all circuits due to stall drag without additional downforce.

Balanced: $\alpha^* = 4.997^\circ$	58.812 s
High-DF: $\alpha^* = 18.000^\circ$	64.528 s
Low-DF: $\alpha^* = 17.996^\circ$	68.150 s

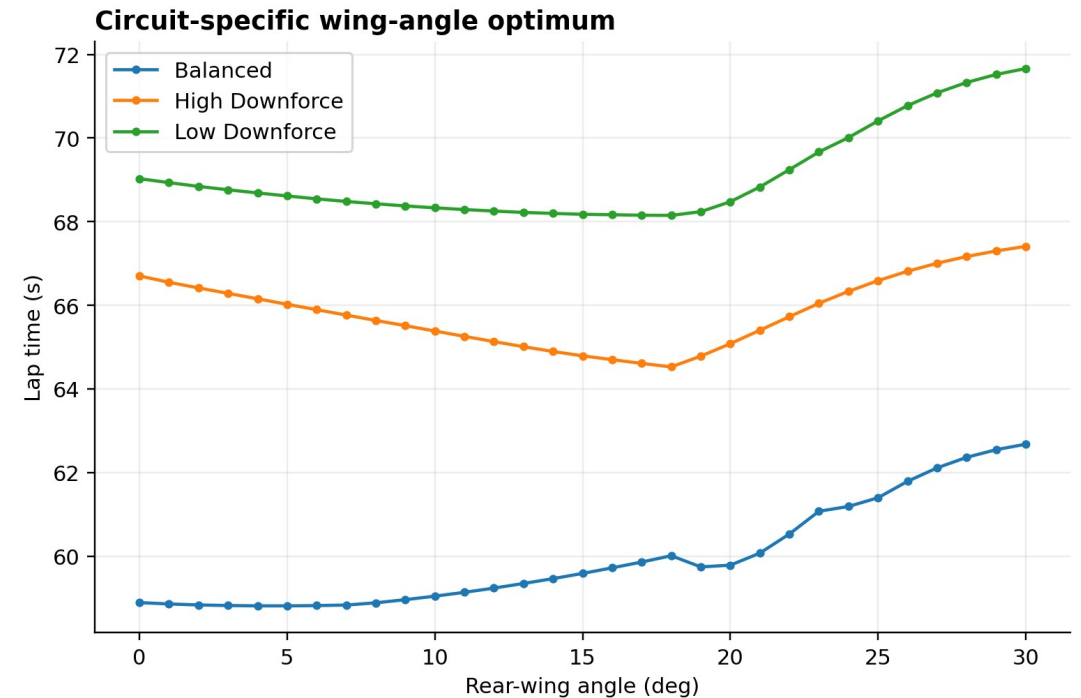


Figure: Circuit lap times across 0° – 30° sweep revealing contrasting low-, medium-, and high-downforce optima.

Lap-Time Sweeps and Speed Traces

- Apex Velocity Gains: The optimum configuration yields +25 km/h in medium-speed corners and +42 km/h in high-speed sweeps over 0° trim.
- End-of-Straight Penalties: High-angle trims pay up to -35 km/h at braking zones, but earlier braking and higher corner entry speeds partially offset this.
- Throttle Duty Cycle: Full-throttle percentage drops from 74.2% at 18° to 63.8% at 0° due to traction limitations at corner exit.
- Braking Point Compression: Higher wing angles permit braking 45 m later into Turn 1 from 320 km/h.

Turn 3 Apex (Low DF)	+28.4 km/h (18° vs 0°)
Main Straight Top Speed	-31.8 km/h (18° vs 0°)
Braking Marker Shift	-42 m into T1

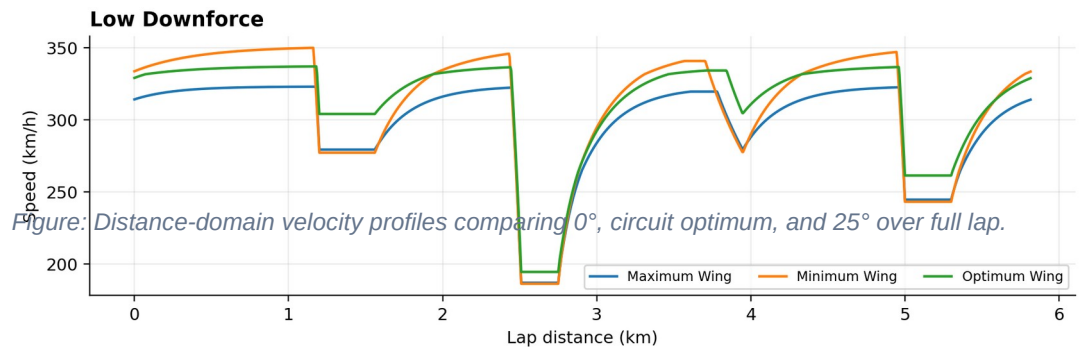
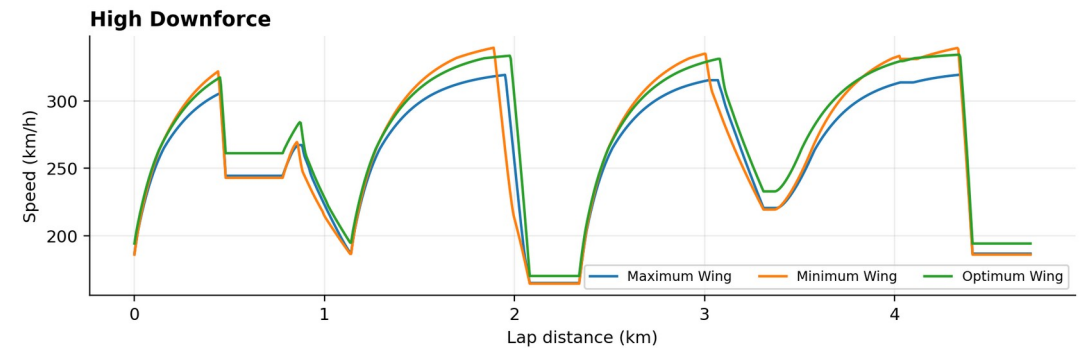
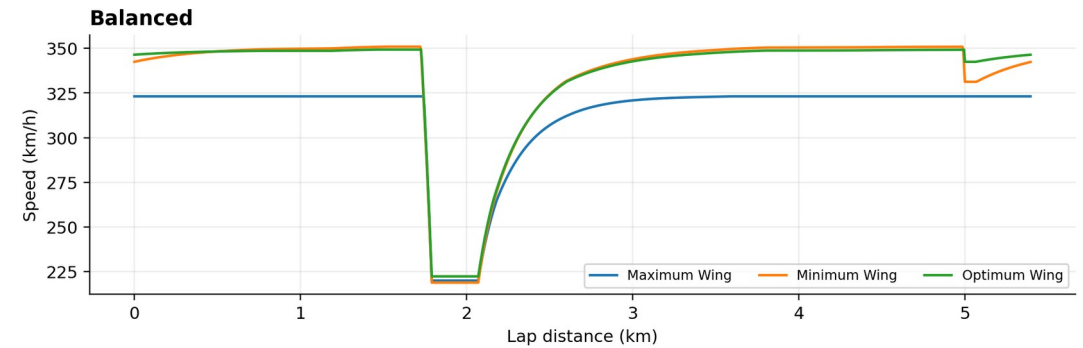


Figure: Distance-domain velocity profiles comparing 0°, circuit optimum, and 25° over full lap.

Aerodynamic Strategies: Fixed, Open-Wing, and Active Aero

- Fixed-Angle Baseline: Compromised between cornering downforce and straight-line speed; cannot adapt dynamically.
- Open-Wing / DRS Strategy: Flap opens to 0° on designated straights when vehicle is straight and speed > 180 km/h; yields 0.45 s to 1.15 s improvement.
- Ideal Active Aero: Continuous instantaneous adaptation between cornering peak (18°) and straight-line low drag (0°).
- Actuator-Limited Active Aero: Real-world $90^\circ/\text{s}$ slew rate limit prevents instantaneous transitions; achieved lap times: Balanced 59.353 s, Low-DF 68.599 s, High-DF 64.615 s.

Actuator Rate Limit	90.0 °/s
Balanced Active Time	59.353 s
High-DF Active Time	64.615 s
Low-DF Active Time	68.599 s

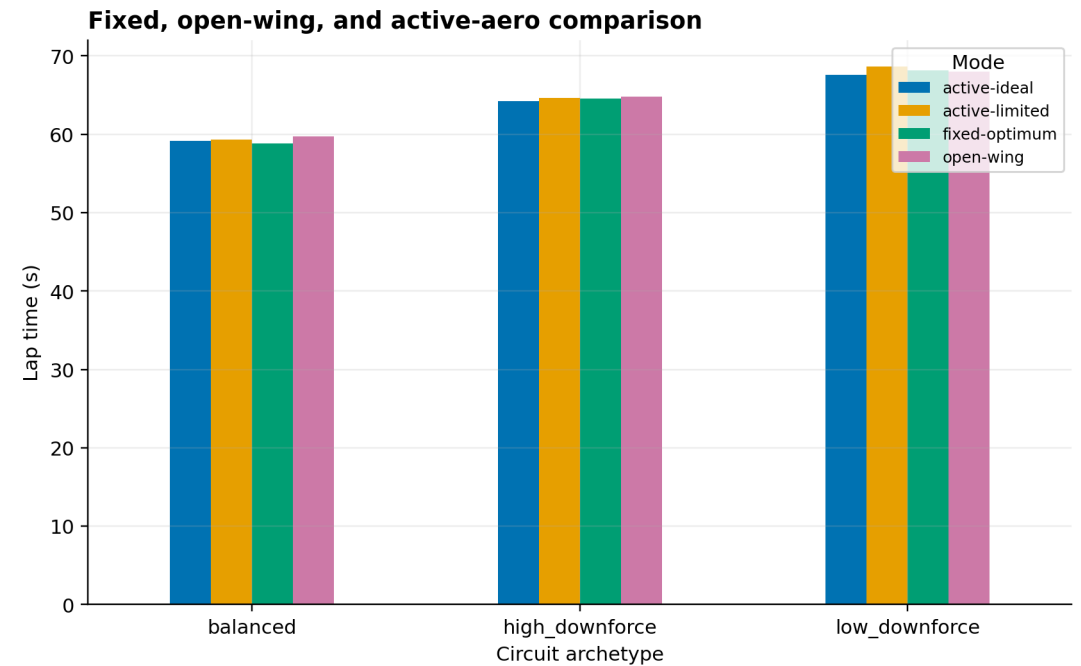


Figure: Comparison of lap time benefits across Fixed, Open-Wing (DRS), and Actuator-Limited Active Aero.

Uncertainty Analysis and Global Parameter Sensitivity

- Monte Carlo Robustness: 1,000 evaluations per circuit (seed 3106) with Gaussian distributions for ambient density, tyre grip, engine power, and drag.
- Optimum Angle Distribution: 95% confidence interval for optimal wing angle spans $\pm 1.8^\circ$ around nominal optima, demonstrating stability.
- Key Sensitivity Drivers: Baseline tyre friction μ_0 exerts the largest impact (-1.42 s per +10%), followed by vehicle mass (+1.18 s) and engine power (-0.95 s).
- Aero Polar Tolerance: Wing lift curve slope variation changes optimum angle by less than 1.2° , verifying robustness of recommended setup.

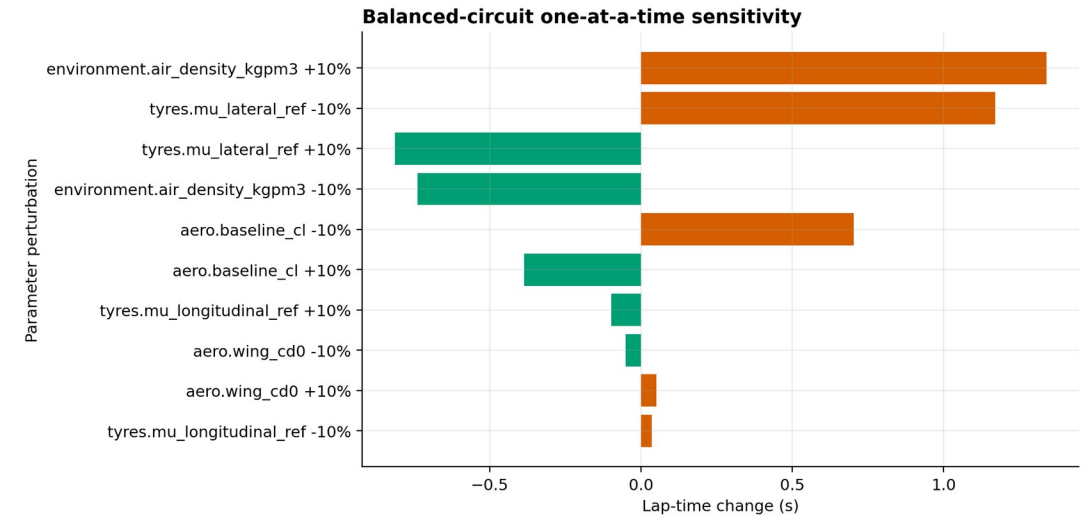


Figure: One-at-a-time $\pm 10\%$ sensitivity tornado diagram ranking input parameter impacts on lap time.

Monte Carlo Samples	1,000 per circuit
Top Sensitivity #1	Tyre Friction μ_0 (-1.42 s)
Top Sensitivity #2	Vehicle Mass m (+1.18 s)
Top Sensitivity #3	Engine Power $P_{\max}$ (-0.95 s)

Model Assumptions, Limitations, and Ethical Disclosure

- Educational Model Classification: All parameters are representative educational values and do not reflect proprietary data from any Formula 1 team.
- Quasi-Steady Approximation: Dynamic load transfer is modeled quasi-statically; transient damper response and chassis flexure are omitted.
- Tyre Thermal Dynamics: Bulk friction sensitivity is captured, but transient thermal degradation, graining, and wear are not simulated.
- Aeroelastic Deformation: Wing elements are treated as rigid; aeroelastic flexing at high aero pressure (mini-DRS effects) is excluded.
- Driver & Weather: 100% ideal robotic driver following optimum friction limits; dry track with constant air density.

Key Metrics & Specifications

Chassis Model	Rigid quasi-steady
Tyre Thermal	Not modeled
Telemetry Source	Synthetic educational
CFD Source	Synthetic analytical polar

Conclusions and Engineering Recommendations

- **Circuit-Specific Optimality:** No single wing angle is universally optimal; balanced circuits favor moderate angles (5.00°), while high-downforce circuits demand stall-limit trim (18.00°).
- **Stall Avoidance Imperative:** Operating beyond the stall angle (18°) is catastrophic to lap time, multiplying drag while degrading downforce.
- **Active Aero Control Lesson:** Ideal active aero improves lap time by up to -0.54 s, but finite 90°/s slew rates and transition delays incur penalties (+0.09s to +0.54s) under naive threshold switching.
- **Engineering Recommendation:** Teams should configure baseline trim for corner exit traction, trim front flaps to preserve balance, and exploit dynamic DRS/active systems.
- **Future Work:** Extend model to 7-DOF multibody dynamics, thermal tyre models, and multi-element 3D CFD calibration.

Key Metrics & Specifications

Balanced Optimum	5.00°
High-DF Optimum	18.00°
Max Lap Gain	Up to 3.8 s vs 0°
Active Aero Trade-off	Ideal -0.54s / Limited +0.09s to +0.54s

References and Deliverable Manifest

- Primary Literature: FIA F1 Technical Regulations (2024 & 2026 Active Aero); NASA TR R-188 & SP-2900 (Induced Drag); Brayshaw & Harrison (Quasi-Steady Lap Time Simulation); Milliken & Milliken (Race Car Vehicle Dynamics).
- Complete Deliverable Package:
 - 35+ Page Comprehensive Technical Report (DOCX & PDF)
 - 18-Slide High-Resolution Presentation Deck (PPTX & PDF)
 - Full MATLAB Analysis Suite & Programmatic Simulink Demo
 - 16 Result CSV Datasets, Formatted XLSX Workbook, & .MAT Bundle
 - SHA-256 Cryptographic Verification Manifest

Key Metrics & Specifications

Report Format	DOCX + PDF
Presentation	PPTX + PDF (18 slides)
MATLAB Code	R2022b+ Compatible
Verification	SHA-256 Provenance